



## Lab 12

# MOSFET Current Source and Current Mirrors

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### 12.1 Objective

To design MOSFET current source and current mirror circuits and utilize current-mirror circuit for biasing of MOSFET CS amplifier.

### Task 1: Pre-Lab

Q1. Why do we use MOSFETs as a Current Source, Explain?

Q2. What is MOSFET Current Mirror? What is the significance of using Current Mirrors?

**Lab 12**  
**MOSFET Current Source and Current Mirrors**

|                                       |                          |
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**12.1 Objective**  
To design MOSFET current source and current mirror circuits and utilize current-mirror circuit for biasing of MOSFET CS amplifier.

**Task 1: Pre-Lab**  
Q1. Why do we use MOSFETs as a Current Source, Explain?

A MOSFET operating in the saturation region (also called active region) exhibits a unique characteristic: the drain current ( $I_d$ ) becomes relatively independent of the drain-source voltage ( $V_{ds}$ ) and depends primarily on the gate-source voltage ( $V_{gs}$ ). This property makes MOSFETs ideal for current source applications. They provide a wide voltage compliance range over which the constant current is maintained, allowing flexible circuit design.

**Advantages of MOSFET Current Sources**

- Voltage Compliance:** Wide range of output voltage over which constant current is maintained
- Tunability:** Current magnitude easily adjusted via  $W/L$  ratio or  $V_{GS}$
- Integration:** Compatible with standard IC fabrication processes
- Low Power:** Negligible gate current in steady-state
- Scalability:** Multiple current sources easily implemented on same chip

Q2. What is MOSFET Current Mirror? What is the significance of using Current Mirrors?

A current mirror is a circuit designed to replicate a reference current ( $I_{ref}$ ) into one or more output branches with high accuracy. The basic two-transistor MOSFET current mirror consists of:  
Input transistor (M1): Diode-connected, establishes reference current. Output transistor (M2): Mirror transistor that replicates the current.

**Fundamental Significance**

- Current Replication:** Provides multiple identical or scaled current sources from single reference
- Essential for biasing multiple stages in analog circuits**
- High Output Impedance:** Acts as near-ideal current source. Minimizes loading effects on connected circuits
- Temperature and Supply Tracking:** All mirrored currents track variations together. Provides inherent compensation in differential circuits

## 12.2 Introduction to Channel Length Modulation

From the ideal characteristics curves of MOSFET, in saturation region drain current is independent of  $V_{DS}$ . Thus, a change in  $V_{DS}$  a zero change in  $I_D$ , which implies that the incremental resistance looking into the drain of a saturated MOSFET is infinite. This, however, is an idealization based on the premise that once the channel is pinched off at the drain end, further increases in  $V_{DS}$  have no effect on the channel's shape. But, in practice, increasing  $V_{DS}$  beyond  $V_{OV}$  does affect the channel somewhat. Specifically, as  $V_{DS}$  is increased, the channel pinch-off point is moved slightly away from the drain, toward the source. This is illustrated in Figure 1, from which we note that the voltage across the channel remains constant at  $V_{OV}$ , and the additional voltage applied to the drain appears as a voltage drop across the narrow depletion region between the end of the channel and the drain region.

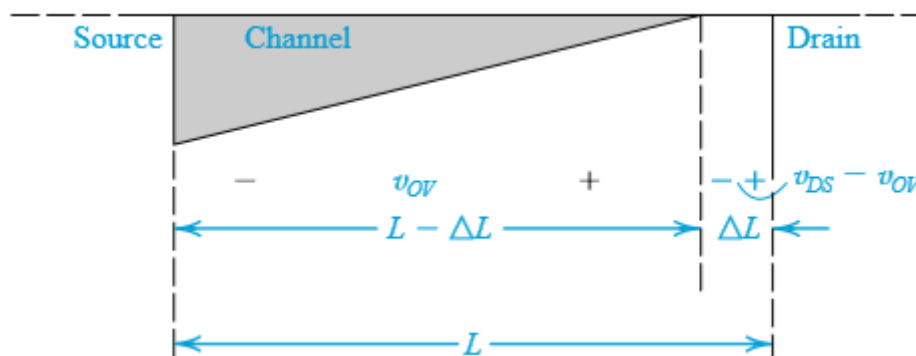


Figure 1: Channel-length modulation

This voltage accelerates the electrons that reach the drain end of the channel and sweeps them across the depletion region into the drain. Note, however, that (with depletion-layer widening) the channel length is in effect reduced, from  $L$  to  $L - \Delta L$ , a phenomenon known as **channel-length modulation**. Now, since  $I_D$  is inversely proportional to the channel length,  $I_D$  increases with  $V_{DS}$ . This effect is accounted in drain current equation in saturation region by including channel-length modulation coefficient  $\lambda$ .

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 (1 + \lambda V_{DS})$$

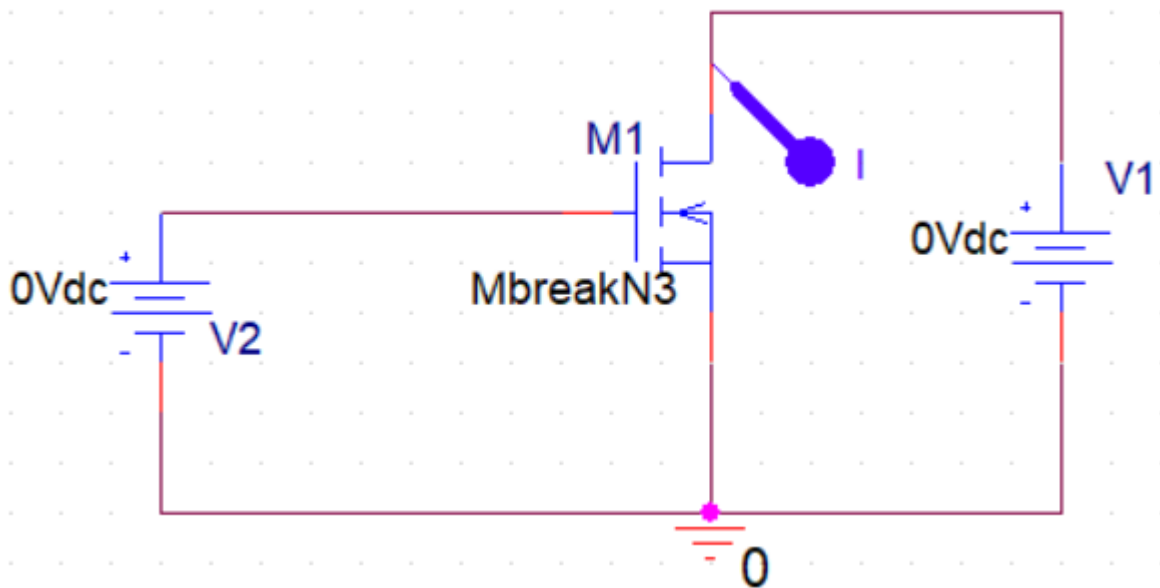
### Task 2: To incorporate channel length modulation in MOSFET PSpice model and obtain output characteristic curves

The channel-length modulation coefficient  $\lambda$  ( $V^{-1}$ ) is modelled as PSpice parameter **LAMBDA** in model editor like other parameters **VTO** (Threshold Voltage), **KP** (Process trans-conductance), **W** (width) and **L** (length).

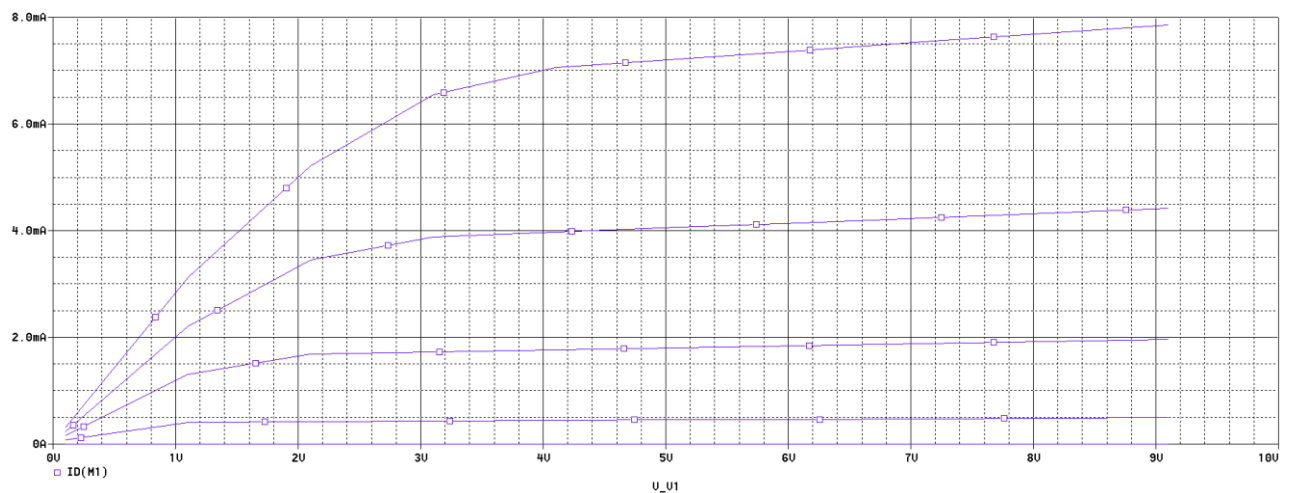
1. Add an MbreakN3 part from PSpice library or use discrete NMOS model. Edit the model using PSpice Model Editor to specify the following given specifications:
  - Threshold voltage: 1V
  - Process trans-conductance: 0.8m A/V<sup>2</sup>
  - Length: 0.8  $\mu$ m
  - Width: 0.8  $\mu$ m

- Channel-length modulation coefficient:  $0.025 \text{ V}^{-1}$
- Obtain characteristic curves ( $I_D$  vs  $V_{DS}$ ) for the modelled MOSFET by sweeping drain to source voltage from 0 to 10V for the different gate-to-source voltages (1 to 5V with step-size of 1V).
  - Add captures of the obtained plot.

**Schematic:**



**Plot**





4. For each of the obtained curves, find the value of output resistance  $r_o$  using the formula given below. Complete Table 1.

$$r_o = \left. \frac{\partial V_{DS}}{\partial I_D} \right|_{V_{GS}=\text{constant}} = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{constant}}$$

Table 1: Effect of channel length modulation on output resistance of MOSFET

| Gate-to-Source Voltage $V_{GS}$ | Output Resistance: $r_o$ |
|---------------------------------|--------------------------|
| 2V                              | 100k                     |
| 3 V                             | 25.839k                  |
| 4 V                             | 11.061k                  |
| 5 V                             | 3.777k                   |

5. How are these curves different from the ones obtained in previous labs? Is there any effect of channel-length modulation on MOS current in saturation region? What is the effect of gate-to-source voltage on output resistance?

Saturation region appears as horizontal lines; these curves show positive slope in saturation region due to channel length modulation. Channel length modulation causes drain current to increase with increasing  $V_{DS}$  in saturation region. This is because as  $V_{DS}$  increases, the effect channel length decreases. This reduces channel resistance. The effect of gate-to-source voltage on output resistance is inverse. As  $V_{GS}$  increases, output resistance decreases.

### 12.3 MOSFET as Current Source

An ideal current source is a two-terminal circuit which provides a constant and stable output current value regardless of the voltage across the output load. The current supplied by such a source is independent of the voltage across source terminals. Real current sources, however, present a slope in their voltage-current characteristics. The value of this slope is given by the conductivity of the **source resistance** placed in parallel with the source. The source resistance is not physically present in the device but it is rather a way to represent the non-ideal output of current source in model. This is shown in Figure 2. The ideal current source has an infinite resistance whereas, for a practical current source, the resistance is not infinite but large in magnitude.

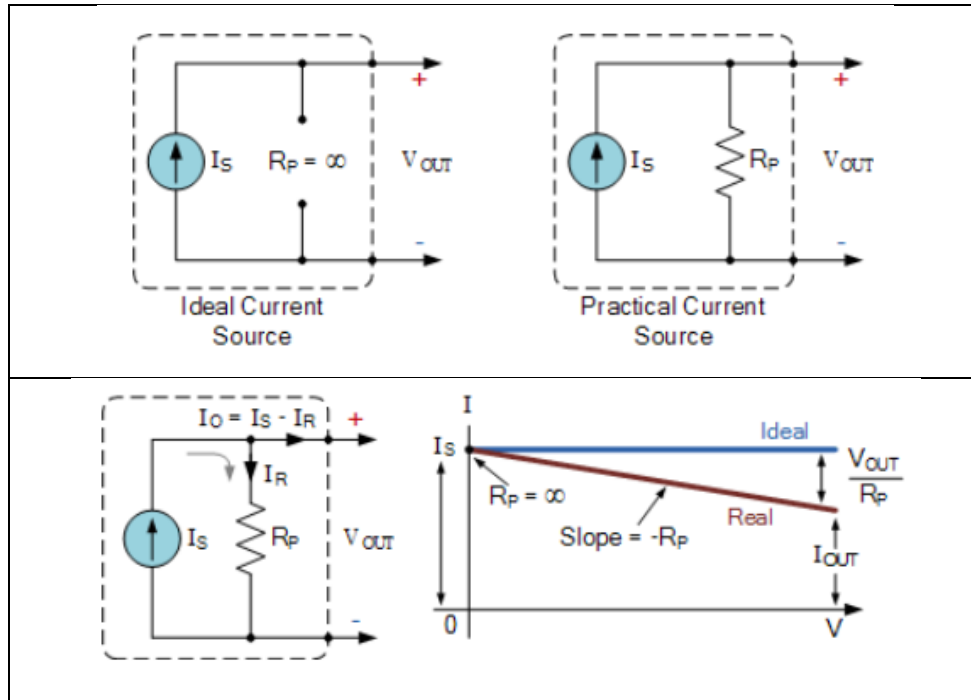


Figure 2: Ideal and practical current sources

From the drain characteristics of MOSFET, we can conclude that a MOSFET can be used as a current source when operated in saturation region. The drain current depends on gate-to-source voltage. Therefore, MOSFET in saturation region acts as voltage-controlled current source.

In the graph of Figure 2, as the load voltage increases, the supply-current of a practical source decreases. When MOSFET is used as a current-source, the load is connected at drain terminal. If load voltage increases due to loading of source, the drain-to-source voltage of MOSFET will decrease. Comparing with the drain-characteristic curve of MOSFET, as drain-to-source voltage decreases, drain current will decrease if channel-length modulation is taken into account i.e. the MOS acts as practical current source. Else, the drain current will remain constant provided the transistor keeps operating in saturation region i.e. the transistor behaves as ideal current source.

The circuit shown in Figure 3 has a diode-connected MOSFET. In previous labs, we have seen that this MOSFET operates in saturation region as its gate and drain terminals are tied together. To design this circuit for any required value of drain current, the required gate-to-source voltage is first calculated using,

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

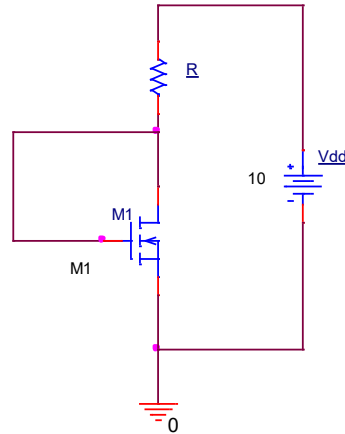


Figure 3: Diode connected MOSFET to establish reference current for current source

As gate-to-source voltage is equal to drain-to-source voltage here, therefore, this is the required drain voltage. The required drain current can be established by using a particular value of resistance  $R$  which is found by using  $V_{DS}=V_{GS}$  in the given KVL equation for drain to source loop.

$$V_D = V_{DD} - I_D R$$

### Task 3: To establish reference current for MOSFET current source

1. Edit the modelled MOSFET in Task 1 to ignore the effect of channel length modulation.
2. When this MOSFET is to be used in circuit of Figure 3, calculate the required gate-to-source voltage and resistance to obtain 1mA drain current. Let,  $V_{DD}$  be 10V.

Table 2: Data-collection table for circuit shown in Figure 3

| Parameters                       | Calculated Values | Simulated Results |
|----------------------------------|-------------------|-------------------|
| Drain Current: $I_D$             | 1mA               | 1mA               |
| Resistance: $R$                  | 7.419kOhm         | 7.42kOhm          |
| Gate-to-Source Voltage: $V_{GS}$ | 2.58V             | 2.581V            |
| Gate-to-Source Voltage: $V_{DS}$ | 2.58V             | 2.581V            |

3. Using the obtained values, simulate the circuit and take measurements for Table 1.

$$I_D = \frac{1}{2} k_n \frac{W}{L} (V_{GS} - V_t)^2$$

$$\sqrt{\frac{2I_D L}{k_n W}} + V_t = V_{GS}$$

$$\sqrt{\frac{2(1m)(0.8u)}{(0.8u)(0.8m)}} + 1 = V_{GS}$$

$$V_{GS} = 2.58V$$

$$V_{DS} = V_{DD} - I_{DR}$$

$$2.58 = 10 - 1m * R$$

$$R = \frac{10 - 2.58}{1m} = 7.419 \text{ k}\Omega$$

## 12.4 Current Mirror Circuits with MOSFETs

The circuit shown in Figure 4 is called current-mirror circuit.

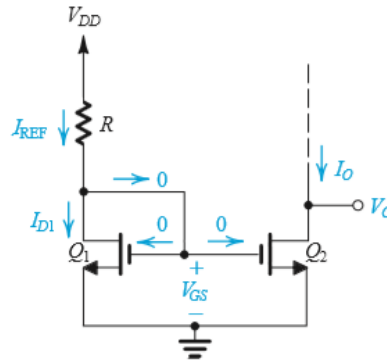


Figure 4: Current-mirror circuit

The heart of this circuit is Q1 transistor through which reference current is established as done in previous task.

$$I_{REF} = \frac{1}{2} k_n' \left( \frac{W}{L} \right)_1 (V_{GS} - V_t)^2$$

Another transistor Q2 is connected such that its gate and source terminals are connected with the gate and source terminals of Q1 transistor. If we assume that Q2 operates in **saturation** mode then, the current  $I_O$  will be

$$I_O = \frac{1}{2} k_n' \left( \frac{W}{L} \right)_2 (V_{GS} - V_t)^2$$

If both the transistors have equal threshold voltage and process trans-conductance parameters, then the output current of Q2 can be related to the reference current by

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

This is a simple and attractive relationship used to calculate current gain or current transfer ratio of this circuit. The special connection of Q1 and Q2 provides an output current  $I_O$  that is related to the reference current  $I_{REF}$  by the ratio of the aspect ratios of the transistors. In the special case of identical transistors  $I_{REF} = I_O$ , and the circuit simply replicates or mirrors the reference current in the output terminal. This



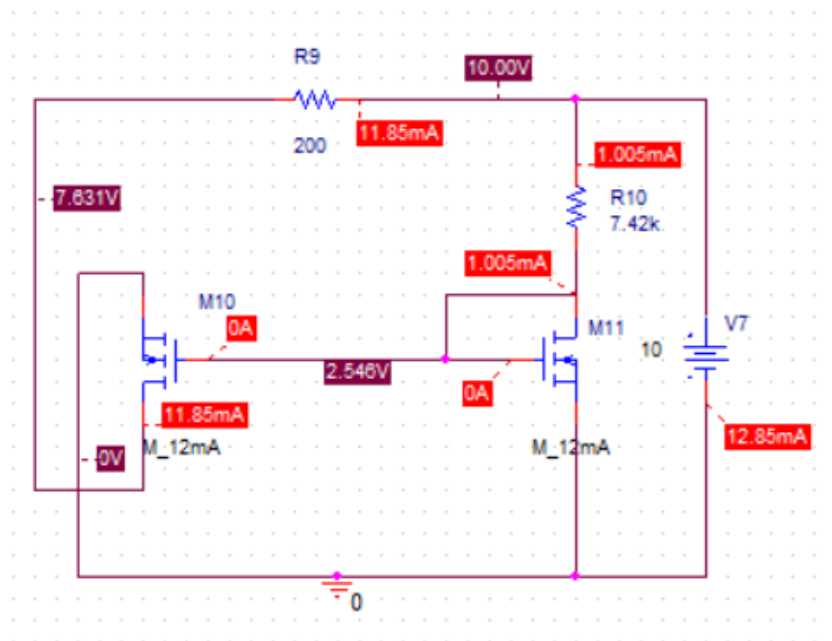
has given this circuit the name of current mirror, a name that is used irrespective of the ratio of device dimensions.

$I_o$  is the current that flows through any load connected at the drain of Q2. This load can be a resistor or any other circuit.

#### Task 4: To design MOSFET current mirror circuits

1. For the current mirror circuit shown in Figure 4, let Q1 be the MOS transistor modelled in previous task. The reference current has already been established i.e. 1mA.
2. Find the required aspect ratio for Q2 transistor that gives output drain current of 12mA.
3. Add another MOS model in PSpice with the parameters required to model Q2 transistor.
4. Modify the circuit implemented in Task 2 to add Q2<sup>1</sup>. Q2 is to be connected with same supply through a load resistor of 200 Ohms.
5. Simulate the circuit and measure the output current. Measure this load current for different values of load resistor.<sup>2</sup>

#### Simulated Circuit:



<sup>1</sup> Do not copy paste the MOSFETs instead add a new MOSFET from component library (Place part).

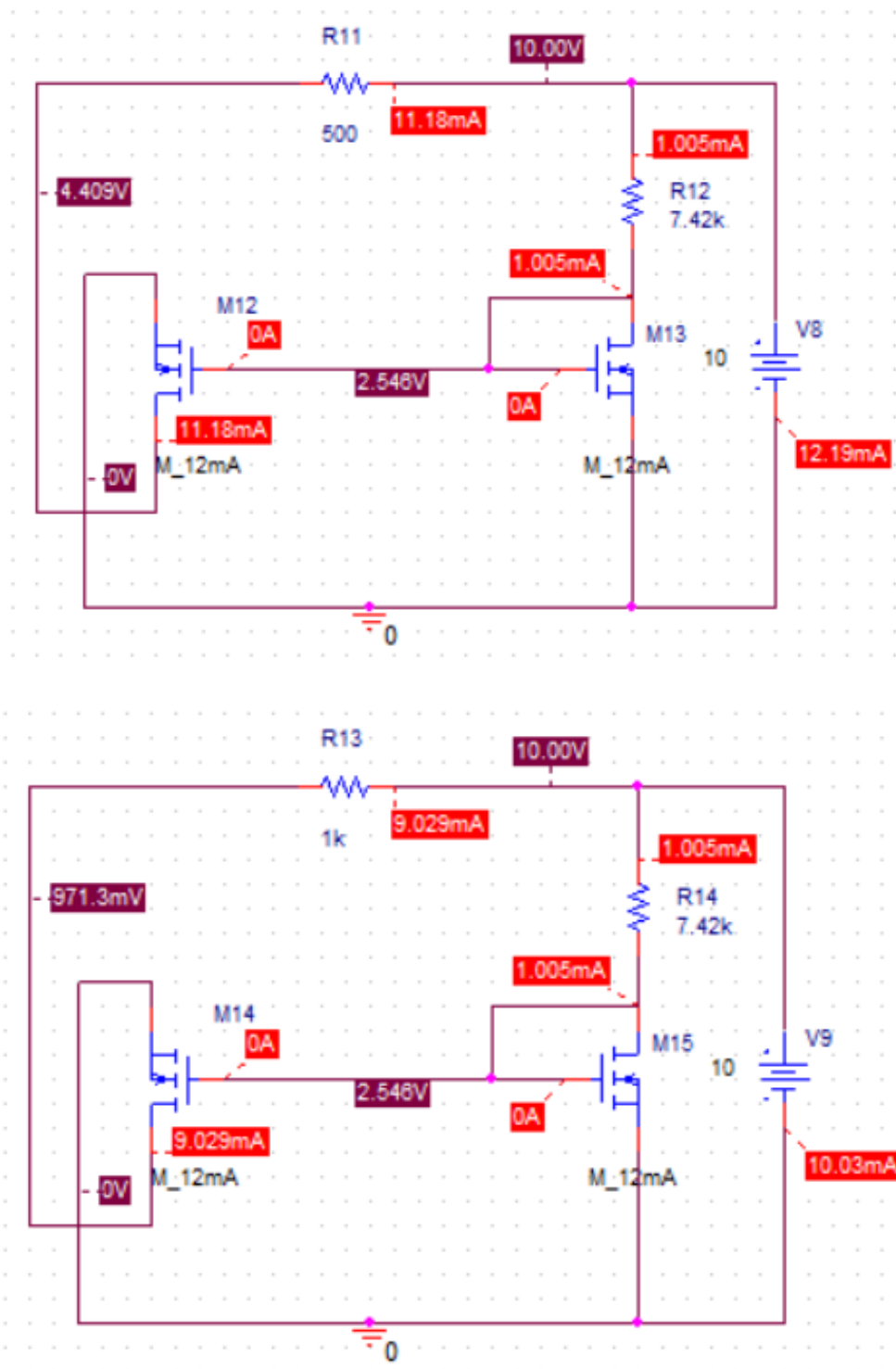


Table 3: Output current of current-mirror circuit at different loads

| Output Current       | Load Resistor<br>(200 Ohms) | Load Resistor<br>(500 Ohms) | Load Resistor<br>(1k Ohms) |
|----------------------|-----------------------------|-----------------------------|----------------------------|
| Drain Current: $I_o$ | 11.85mA                     | 11.18mA                     | 9.029mA                    |

6. What is the maximum value of load resistor for which the transistor Q2 keeps operating in saturation region?

We know that  $V_{DS}$  must be greater than or equal to  $1.6V$

Using the values:

- $V_{DD} = 10V$
- $I_O = 11.85mA$
- $V_{GS} - V_t = 1.546V$

$$R_L = \frac{10 - 1.546}{0.01185}$$
$$R_L = \frac{8.454}{0.01185}$$
$$R_L \approx 713.417\Omega$$

7. Does this circuit provide the required constant current of  $12mA$  to the load? Refer the results collected in last Table and use the maximum value of load resistor to interpret the required condition for correct working of MOS current-mirror circuit?

At  $200\Omega$  and  $500\Omega$  the result is almost accurate, however at  $1k\Omega$  it gives  $9.020mA$ . For current mirror to work properly both transistors need to be in saturation region. We can keep the load resistance below critical threshold. If the load goes above  $713.417\Omega$ , transistor enters linear region, causing current mirror to fail.

## 12.5 Biasing MOSFET CS Amplifier by Current Source

In previous labs, we have been using voltage-divider circuit for biasing of CS amplifier. In this lab, we will bias CS amplifier circuit of Lab 10 using a current source. Remember, biasing is needed to establish certain operating point ( $I_D$  and  $V_D$ ) for MOSFET and for its application as an amplifier; it needs to be biased in saturation region. Using constant current source is the most efficient method by which a MOSFET can be biased. A general representation for biasing of MOSFET using current source is shown in Figure 5.

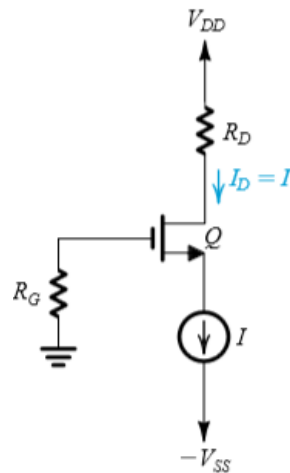


Figure 5: Biasing by constant current source

The current-mirror circuit implemented in Task 4 using MOSFETs can be used as current source for MOSFET biasing. Here,  $R_G$  resistor (usually in range of  $M\Omega$ ) establishes a DC ground at the gate and presents a large input resistance to the input signal source which can be capacitively coupled to the gate. The complete CS amplifier circuit with coupling capacitors and by-pass is shown in Figure 6. The current source connected here for biasing the transistor is shown in Figure 7. The current mirror utilizes positive and negative sources ( $V_{DD}$  and  $-V_{SS}$ ).

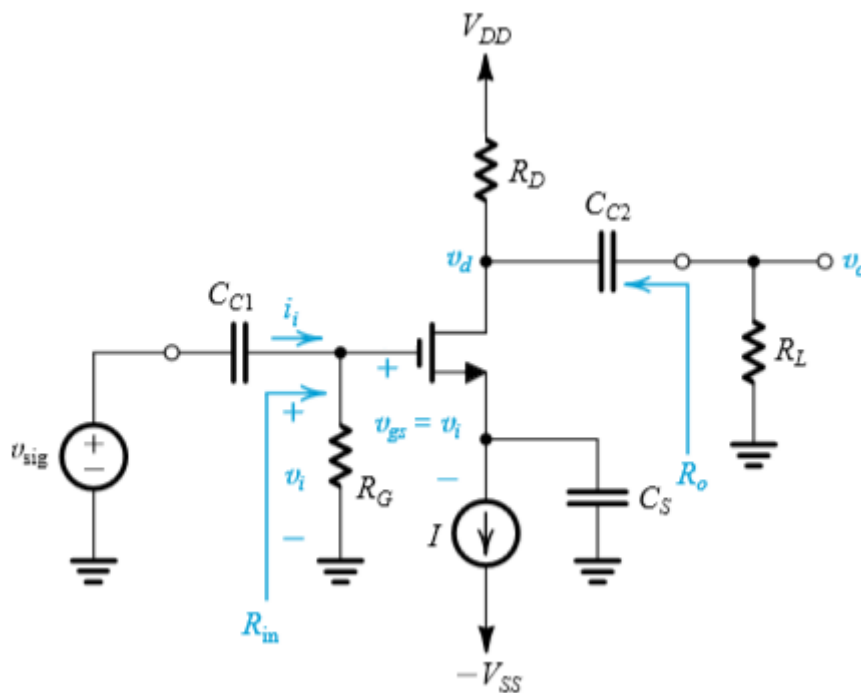


Figure 6: Common-source amplifier circuit with constant current source biasing

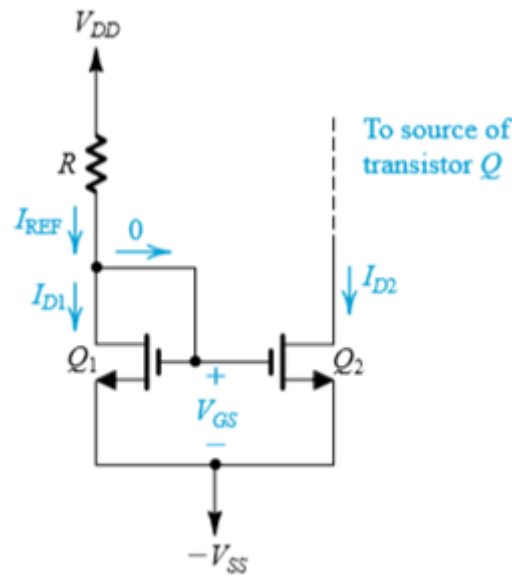


Figure 7: Constant-current source circuit to be used for CS amplifier biasing

### Task 5: To design and implement CS amplifier circuit biased by current source

1. Implement the CS amplifier circuit shown in Figure 6 using 2N7000 MOSFET. Let,  $R_G=1\text{Mega}\Omega$ ,  $R_D=220\Omega$ ,  $V_{DD}=5\text{V}$ ,  $V_{SS}=-5\text{V}$ ,  $C_1=C_2=10\mu\text{F}$  and  $C_s=15\mu\text{F}$ .
2. In place of current-source, connect the output terminal (drain terminal) of a current-mirror circuit (Figure 7) that establishes drain current of 12mA for the MOS amplifier.
3. Apply AC signal of 10mV (amplitude) at 5 kHz frequency at the input of this amplifier and measure unloaded output voltage. (To simulate open circuit, connect 100 Mega $\Omega$  in place of load resistor  $R_L$ ).
4. Complete table 3 to characterize the implemented CS amplifier

### Schematic:

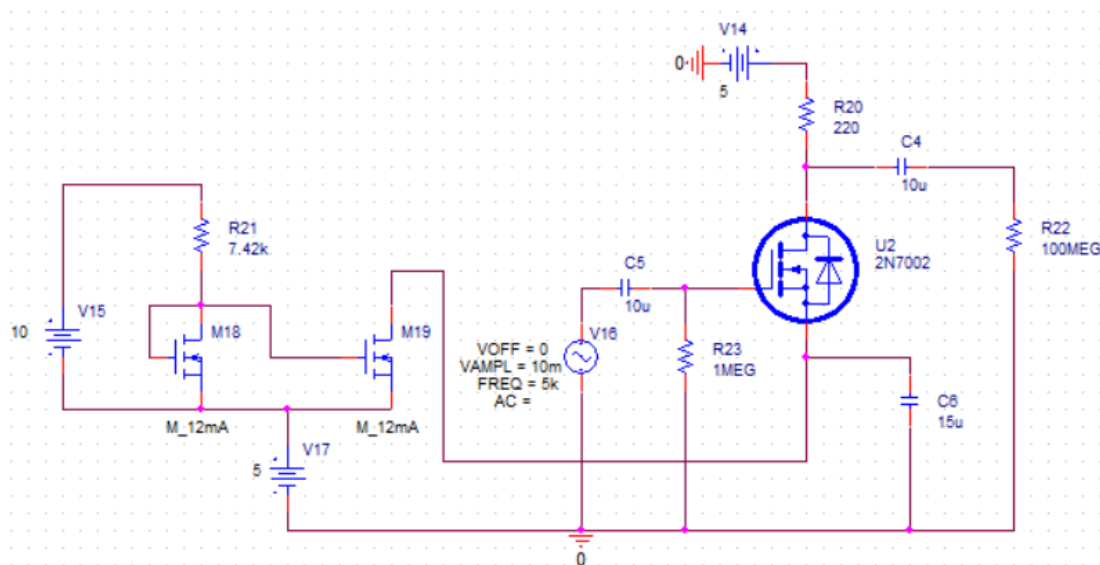


Table 3: CS Amplifier biased by current source

| Parameters                                           | Values     |
|------------------------------------------------------|------------|
| Drain Current: $I_D$                                 | 1.01m      |
| Drain Voltage: $V_D$                                 | 4.75       |
| Source Voltage: $V_S$                                | -2.21      |
| Drain-to-Source Voltage: $V_{DS}$                    | 6.95       |
| Gate-to-Source Voltage: $V_{GS}$                     | 2.213      |
| Trans-conductance: $g_m = \frac{2I_D}{V_{GS} - V_T}$ | 0.000926   |
| Calculated Gain: $A_v = -g_m R_D$                    | -0.2       |
| Measured Gain: $A_v$                                 | 478        |
| Mode of Operation of Q (2N7000)                      | Saturation |

## Task 6: Post Lab

Q1. How can we improve the performance of a current mirror circuit?

**ANSWER:**

The performance of a current mirror can be improved by increasing output resistance using configurations like the cascode or Wilson mirror, which minimize the effect of channel-length modulation and keep the output current constant despite changes in  $V_{DS}$ . Better matching between transistors, through larger device areas, common-centroid layout, and using identical  $W/L$  ratios, enhances current copying accuracy. Adding source degeneration resistors improves thermal stability and reduces sensitivity to threshold voltage mismatches, while using longer channel lengths lowers the  $\lambda$  effect. For applications requiring very high precision, regulated cascode mirrors with feedback amplifiers can be employed.

Q2. What is Wilson Current Mirror? Which type of current mirror is used most?

**ANSWER:**

A Wilson current mirror is an improved current mirror circuit that uses three transistors (instead of two) to provide higher output resistance and better current matching. It introduces negative feedback: the output transistor's drain voltage is sensed and used to correct the gate voltage, making the output current much less sensitive to variations in  $V_{DS}$  and reducing the error caused by base currents (in BJTs) or



channel-length modulation (in MOSFETs). Compared to the simple two-transistor mirror, the Wilson mirror offers significantly flatter output characteristics and higher accuracy over a wider voltage range. The cascode current mirror is most commonly used in high-performance analog integrated circuits because it achieves very high output resistance with minimal extra complexity, while the simple current mirror remains the most widely used overall due to its simplicity and low voltage headroom requirement. The choice depends on the specific application: simple mirrors for low-cost, high-headroom designs; cascode or Wilson mirrors for precision analog circuits like operational amplifiers and data converters.



## 1.1 Assessment Rubric

### Lab 12 MOSFET Current Source and Current Mirrors

|                                                                    |                                       |
|--------------------------------------------------------------------|---------------------------------------|
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#### Points Distribution

| Task No.              | LR2<br>Simulation<br>32 | LR 4<br>Data Collection<br>20 | LR 5<br>Plots<br>8 | LR 6<br>Calculation<br>8 | LR 10<br>Analysis<br>24 |
|-----------------------|-------------------------|-------------------------------|--------------------|--------------------------|-------------------------|
| Task 1                |                         |                               |                    |                          | /8                      |
| Task 2                | /8                      | /6                            | /4                 | -                        | /4                      |
| Task 3                | /8                      | /4                            | -                  | /4                       | -                       |
| Task 4                | /8                      | /4                            | -                  | /4                       | /8                      |
| Task 5                | /8                      | /6                            | /4                 | -                        | /4                      |
| Task 6                |                         |                               |                    |                          | /4                      |
|                       | /32                     | /20                           | /8                 | /8                       | /28                     |
| <b>CLO<br/>Mapped</b> | CLO 2                   | CLO 1                         | CLO 1              | CLO 2                    | CLO 1                   |

| Affective Domain Rubric |                   | Points | CLO<br>Mapped |
|-------------------------|-------------------|--------|---------------|
| AR 4                    | Report Submission | /4     | CLO 4         |

| CLO          | Total Points | Points Obtained |
|--------------|--------------|-----------------|
| 1            | 56           |                 |
| 2            | 40           |                 |
| 4            | 04           |                 |
| <b>Total</b> | <b>100</b>   |                 |

*For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.*